Name_____ Date_____ Period_____

LESSON 4-1 · HOMEWORK PACKET · LEVEL 1 · SHEET 1 OF 2

Percent Means Per Hundred

Maryland Standard 6.AT.4 · Learning target – I can write a ratio as a percent by finding the equivalent ratio out of 100.

REMEMBER

- **Percent** means "per hundred." 37% = 37 out of 100.
- On a hundred grid, each square is **1%**.
- Make an **equivalent ratio** out of 100: multiply **both** terms by the same number.

WORKED EXAMPLE

Out of every 20 students, 7 ride the bus. What percent of the students ride the bus?

Step 1 20 × 5 = 100, so multiply both terms by 5.

Step 2 7 × 5 = 35, so 7 out of 20 = 35 out of 100.

Step 3 35 out of 100 = 35%.

Answer **35%**

WORD BANK

percent (por ciento) – out of 100

% (signo de por ciento) – the percent sign

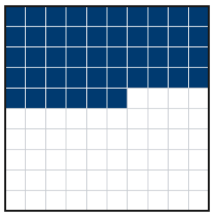
whole (el entero) – all of it; the whole is 100%

equivalent ratio (razón equivalente) – the same comparison with new numbers

PRACTICE

show your work · label units

1. The grid has 100 equal squares.



Step 1 Count the shaded squares. Each square is 1%.

Step 2 Subtract from 100% to find the part not shaded.

a. What percent of the grid is shaded? _____ %

b. What percent is **not** shaded? _____ %

2. Write each ratio as a percent. First scale it to an equivalent ratio out of 100.

Step 1 Ask: what times the second number makes 100?

Step 2 Multiply the first number by that same number.

a. 3 out of 10 = _____ %

b. 7 out of 25 = _____ %

c. 1 out of 2 = _____ %

d. 17 out of 50 = _____ %

e. 2 out of 5 = _____ %

f. 3 out of 20 = _____ %

g. 13 out of 25 = _____ %

h. 9 out of 10 = _____ %

3. Animal Shelter. Out of every 25 dogs at a shelter, 9 are puppies. Complete the ratio table. Then write the percent of dogs that are puppies.

Frame *___ puppies out of 100 dogs, so ___% are puppies.*

Step 1 Each new column adds 9 puppies and 25 dogs.

Step 2 Fill the empty boxes. Stop at 100 dogs.

Puppies	9	18		
All dogs	25	50	75	100

Percent of puppies_____ %

4. Garden Plants. A garden has **50 plants**. **22** of them are tomato plants.

Step 1 50 × 2 = 100. Multiply 22 by 2 too.

Step 2 Subtract your percent from 100%.

a. What percent are tomato plants? _____ %

b. What percent are **not** tomato plants? _____ %

5. Find the Error. To write 7 out of 20 as a percent, Ana added 80 to both numbers. She got 87 out of 100, so she wrote 87%. Explain her mistake and find the correct percent.

Frame *Ana's mistake is ___. 20 × ___ = 100, so 7 × ___ = ___, which is ___%.*

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Find a percent on a label, sign, or screen at home. Tell someone what it means "out of 100."

Name_____ Date_____ Period_____

LESSON 4-1 · HOMEWORK PACKET · LEVEL 1 · SHEET 2 OF 2

Percents Greater Than 100%

Maryland Standard 6.AT.4 · Learning target – I can use percents greater than 100% to describe an amount larger than the whole.

REMEMBER

- **100%** is one whole. Over 100% is **more than the whole**.
- 200% = 2 times the whole. 150% = 1.5 times.
- "Times as much" to percent: multiply by 100. 2.5 times = 250%.

WORKED EXAMPLE

A puppy grew from 10 lb to 13 lb. Write 13 lb as a percent of 10 lb (the whole).

Step 1 Scale by 10: 13 to 10 = 130 to 100.

Step 2 130 out of 100 = 130%, more than the whole.

Answer **130%**

WORD BANK

whole (el entero) – the amount that is 100%

more than 100% (más de 100%) – more than the whole

times as much (veces tanto como) – 2 times = 200%, 3 times = 300%

PRACTICE

show your work · label units

1. Write each amount as a percent of the whole.

Step 1 1 time as much is 100%.

Step 2 Multiply the number of times by 100%.

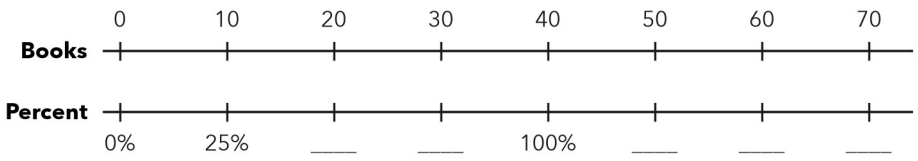
a. 3 times as much = ____ %

b. 1.75 times as much = ____ %

2. **Library Books.** Monday (40 books) is the whole. Tuesday: 70 books. Fill in the double number line. Write Tuesday as a percent of Monday.

Step 1 40 books is 100%, so 10 books is 25%.

Step 2 Count up by 25% for each 10 books.



Tuesday_____ %

3. Bake Sale. A class set a goal of $200. They raised $260. What percent of the goal did they raise?

Step 1 $200 is the whole, 100%.

Step 2 Divide both by 2: 260 to 200 is the same as ___ to 100.

Answer _____ %

4. Chess Club. Last year a chess club had 20 members (the whole). This year it has 33 members. Write this year's members as a percent of last year's.

Frame *33 to 20 = ___ to 100, so this year is ___% of last year.*

Step 1 Last year, 20 members, is the whole, 100%.

Step 2 20 × 5 = 100. Multiply 33 by 5 too.

Answer _____ %

5. Find the Error. Malik says 60 cm is 60% of 30 cm. Explain his mistake and fix it.

Frame *Malik is wrong because 60 is ___ times 30, so 60 cm is ___% of 30 cm.*

_____

_____

TEST PRACTICE

choose the best answer

6. 45 points is what percent of 30 points?

Step 1 30 points is the whole, 100%.

Step 2 Find 45 ÷ 30. Then multiply by 100%.

A 15% **B** 45% **C** 115% **D** 150%

7. 250% of an amount is how many times the amount?

Step 1 100% is 1 time the amount. 200% is 2 times.

Step 2 Divide the percent by 100 to find how many times.

A 0.25 times **B** 2.5 times **C** 25 times **D** 250 times

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Tell someone at home what 200% of their age would be.